

Human Intelligence Layer for Artificial Intelligence: A Decentralized Consensus Economy for OCR, Data Annotation, AI Training and Human-AI Collaboration on Solana

Artur Gi.zycki

June 12, 2026

Abstract

We introduce the Human Intelligence Layer (HIL), a decentralized infrastructure for artificial intelligence training, verification and dataset generation. The proposed architecture combines blockchain technology, consensus mechanisms, machine learning and game-theoretic incentives into a unified economic framework.

Unlike conventional blockchain systems whose security relies on computational work or financial stake, the proposed system rewards participants for producing useful information. Human contributions are aggregated through weighted consensus mechanisms and become training data for continuously improving artificial intelligence models.

The framework is developed initially for Optical Character Recognition (OCR), but naturally extends to multimodal annotation tasks including document processing, speech recognition, image understanding, machine translation and knowledge extraction.

We provide mathematical models of consensus formation, reputation dynamics, economic incentives, dataset valuation, token economics, anti-fraud mechanisms and long-term ecosystem growth. Furthermore, we argue that Solana currently provides the most suitable blockchain infrastructure for such a system due to its high throughput, low transaction costs and scalability characteristics.

Contents

1	Introduction	2
1.1	The Data Bottleneck of Artificial Intelligence	2
1.2	Limitations of Current Annotation Systems	2
2	Conceptual Architecture	3
2.1	Definition of Human Intelligence Layer	3
2.2	Comparison with Existing Systems	3
3	OCR as the Initial Market	3
3.1	Task Structure	3
3.2	Consensus Formation	3
3.3	Theorem: Consistency of Weighted Consensus	4
4	Information Theory of Human Consensus	4
4.1	Entropy	4
4.2	Interpretation	4
4.3	Information Value	4
5	Reputation Dynamics	4
5.1	Theorem	4

6	Game Theory of Participation	5
6.1	Utility Function	5
6.2	Honest Strategy	5
7	Artificial Intelligence Feedback Loop	5
7.1	Theorem	5
8	Economics of Dataset Ownership	5
8.1	Data as Capital	5
9	Why Solana?	6
9.1	Requirements	6
9.2	Comparison	6
9.3	Economic Argument	6
10	Anti-Fraud Theory	6
10.1	Sybil Attacks	6
10.2	Collusion	6
11	Token Economics	6
11.1	Revenue Streams	7
12	Monetization Mechanisms	7
12.1	OCR-as-a-Service	7
12.2	Dataset Licensing	7
12.3	AI Marketplace	7
12.4	Total Revenue	7
13	Growth Dynamics	7
14	Future Extensions	7
15	Conclusion	8

1 Introduction

1.1 The Data Bottleneck of Artificial Intelligence

Modern machine learning systems depend critically upon large-scale labeled datasets.

Let

$$D = (x_i, y_i)_{i=1}^N$$

denote a supervised learning dataset.

The optimization problem can be written as

$$\theta^* = \arg \min_{\theta} L(\theta; D)$$

While computing power continues to increase, high-quality labeled datasets remain expensive to produce.

This creates a fundamental bottleneck for the future development of artificial intelligence.

1.2 Limitations of Current Annotation Systems

Existing annotation systems suffer from:

1. Centralization.

2. High operational costs.
3. Limited scalability.
4. Absence of ownership rights for contributors.
5. Weak incentive structures.

The Human Intelligence Layer attempts to overcome these limitations.

2 Conceptual Architecture

2.1 Definition of Human Intelligence Layer

Define

$$\mathcal{H} = (\mathcal{U}, \mathcal{T}, \mathcal{D}, \mathcal{M}, \mathcal{E})$$

where

- $\mathcal{U}$ denotes users,
- $\mathcal{T}$ denotes tasks,
- $\mathcal{D}$ denotes datasets,
- $\mathcal{M}$ denotes AI models,
- $\mathcal{E}$ denotes economic mechanisms.

The entire ecosystem operates through the cycle

$$\mathcal{U} \rightarrow \mathcal{D} \rightarrow \mathcal{M} \rightarrow \mathcal{E} \rightarrow \mathcal{U}.$$

2.2 Comparison with Existing Systems

Compare:

System	Output	Ownership
Bitcoin	Hashes	Miners
Ethereum	Smart contracts	Validators
OpenAI	Models	Corporation
HIL	Datasets + Models	Network

Super — ta wersja powinna już działać poprawnie.

3 OCR as the Initial Market

3.1 Task Structure

Each OCR task is represented by $\tau = (I, T, u, t)$.

3.2 Consensus Formation

Suppose

$$T_1, \dots, T_n$$

are transcriptions.

Define

$$C(T) = \sum_{i: T_i = T} q_i.$$

Then

$$T^* = \arg \max_T C(T).$$

3.3 Theorem: Consistency of Weighted Consensus

Theorem.

Assume that user accuracy satisfies

$$p_i > \frac{1}{2}$$

for all sufficiently reputable users.

Then

$$\Pr(T^* = T_{true}) \rightarrow 1$$

as

$$n \rightarrow \infty.$$

Sketch of proof.

The result follows from the weighted law of large numbers.

Since each participant has probability exceeding one half of providing the correct answer, the expected weighted vote of the correct label dominates the expected weighted vote of any incorrect label.

Concentration inequalities imply convergence toward the correct consensus.

4 Information Theory of Human Consensus

4.1 Entropy

Define

$$H(I) = -\sum_T p_T \log p_T.$$

4.2 Interpretation

Low entropy:

$$H(I) \approx 0$$

means trivial examples.

High entropy:

$$H(I) \gg 0$$

means difficult examples.

4.3 Information Value

Define

$$V_I = \alpha H(I).$$

The reward mechanism becomes

$$R_i = R_0 + V_I.$$

5 Reputation Dynamics

Define

$$q_i(t+1) = (1-\lambda)q_i(t) + \lambda S(T_i, T^*).$$

5.1 Theorem

The reputation process converges.

Sketch.

The mapping

$$F(q) = (1-\lambda)q + \lambda s$$

is a contraction.

Banach's fixed point theorem implies convergence.

6 Game Theory of Participation

6.1 Utility Function

User utility:

$$U_i = R_i - C_i$$

6.2 Honest Strategy

Let

h_i

denote honest behavior.

Theorem.

Under suitable reward scaling,

h_i

is a Nash equilibrium.

Sketch.

Any deviation decreases expected consensus probability and therefore expected reward.

Thus

$$E[R_i(h_i)] > E[R_i(d_i)].$$

7 Artificial Intelligence Feedback Loop

Dataset:

$$D_n = \{(I_k, T_k)\}_{k=1}^n.$$

Training:

$$\theta_{n+1} = \theta_n \eta \nabla L(\theta_n; D_n).$$

7.1 Theorem

Dataset growth improves expected model quality.

Sketch.

Apply standard PAC-learning bounds.

Generalization error satisfies

$$\epsilon = O(\sqrt{\frac{d}{N}}).$$

Therefore larger datasets improve performance.

8 Economics of Dataset Ownership

Dataset value: $P(D) = aN + bQ + cM$, where N = dataset size, Q = dataset quality, M = market demand.

8.1 Data as Capital

Traditional economy:

Capital $\rightarrow$ *Production*.

HIL economy:

Data $\rightarrow$ *Models* $\rightarrow$ *Revenue*.

9 Why Solana?

9.1 Requirements

The system requires:

1. Millions of users.
2. Billions of daily interactions.
3. Extremely low fees.
4. Fast finality.
5. Mobile support.

9.2 Comparison

Blockchain	Fees	Suitability
Bitcoin	High	Poor
Ethereum	High	Moderate
Cardano	Moderate	Good
Polygon	Low	Good
Solana	Very Low	Excellent

9.3 Economic Argument

Suppose 10^8 tasks occur daily. If average fee equals 0.01 USD then operational costs become 10^6 USD/day.

The economics become impossible.

Therefore extremely low transaction costs are essential.

10 Anti-Fraud Theory

10.1 Sybil Attacks

Assume an attacker creates k identities. Reputation initialization is $q_0 \ll 1$. Thus influence scales as kq_0

The attack remains economically inefficient.

10.2 Collusion

Suppose a coalition size m attempts manipulation. If $m < M_{critical}$ then weighted consensus remains stable.

11 Token Economics

Supply: $S(t)$. Demand: $D(t)$. Token price: $P(t) = \frac{D(t)}{S(t)}$.

11.1 Revenue Streams

1. OCR API.
2. Human verification.
3. Dataset licensing.
4. AI APIs.
5. Enterprise annotation.
6. Historical archive digitization.
7. Government document processing.
8. Medical record processing.

12 Monetization Mechanisms

12.1 OCR-as-a-Service

Revenue: $R_{OCR} = cN_{documents}$.

12.2 Dataset Licensing

Revenue: $R_{Data} = \sum_i P_i$

12.3 AI Marketplace

Revenue: $R_{AI} = \sum_j U_j$

12.4 Total Revenue

$R = R_{OCR} + R_{Data} + R_{AI}$.

13 Growth Dynamics

Define $N(t)$ = users, $D(t)$ = dataset size, $M(t)$ = model quality, $R(t)$ = revenue.

The ecosystem dynamics satisfy $\frac{dN}{dt} = aR - bN$, $\frac{dD}{dt} = cN$, $\frac{dM}{dt} = dD$, $\frac{dR}{dt} = eM$.

This forms a nonlinear positive feedback system.

14 Future Extensions

1. Speech recognition.
2. Translation.
3. Video annotation.
4. Scientific OCR.
5. Mathematical OCR.
6. Legal AI.
7. Medical AI.

8. Federated learning.
9. Autonomous AI agents.
10. Human-AI hybrid governance.

15 Conclusion

The Human Intelligence Layer transforms human cognitive effort into a productive economic asset.

Rather than wasting resources on arbitrary computations, the system converts collective human intelligence into datasets, models and services.

The resulting architecture creates a self-reinforcing cycle of data generation, model improvement and economic growth.

In this sense the Human Intelligence Layer may be interpreted as a decentralized knowledge production engine for the age of artificial intelligence.

References

- [1] Goodfellow, Bengio, Courville. Deep Learning. MIT Press.
- [2] Bishop. Pattern Recognition and Machine Learning. Springer.
- [3] Vapnik. Statistical Learning Theory.
- [4] Shannon. A Mathematical Theory of Communication.
- [5] Nakamoto. Bitcoin.
- [6] Yakovenko. Solana Whitepaper.
- [7] Osborne, Rubinstein. A Course in Game Theory.
- [8] Mohri, Rostamizadeh, Talwalkar. Foundations of Machine Learning.